

The Effect of School Capital Investments: A State-Level Perspective*

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Abstract

We revisit the analysis of Biasi, Lafortune and Schönholzer (2025), published in the *Quarterly Journal of Economics*, Vol. 140, Issue 3, August 2025, which studies the effect of school capital investments on student outcomes and house prices using a rich dataset of school capital projects across the U.S. from 1995–2017. We replicate their main results and extend their analysis by exploring state-level heterogeneity in the treatment effect estimates. We find that the effect of school capital investments varies significantly across states, suggesting that pre-existing differences in school infrastructure, policy implementation, and local economic conditions may play a crucial role in mediating the impact of such investments.

Keywords: School Capital Investments, Student Outcomes, House Prices, State-Level Heterogeneity, STATA, R

JEL Codes: H75, I22, R53

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1 Introduction

Capital investments in schools, such as building new facilities or renovating existing ones, are often justified on the grounds that they can improve student outcomes and increase neighborhood quality. However, the empirical evidence on the effectiveness of such investments is mixed. Seminal work by Cellini, Ferreira and Rothstein (2010) and follow-up work by Martorell, Stange and McFarlin Jr (2016) does not find strong evidence that school construction projects lead to improvements in student achievement in California, whereas other studies, such as Neilson and Zimmerman (2014), Abott et al. (2020) and Baron (2022), find large positive effects on test scores and home prices. More recently, Biasi, Lafortune and Schönholzer (2025) have conducted a comprehensive analysis using a rich dataset of school capital projects across the U.S. from 1995-2017. They find that certain school capital investments have a positive effect on student outcomes, particularly for low-income students. Using an empirical strategy that improves upon prior work by Cellini, Ferreira and Rothstein (2010), they develop a stacked dynamic regression discontinuity (DRD) estimator that leverages discontinuities in bond authorization passage and addresses endogeneity concerns related to the timing of elections and “forbidden controls”. Their findings suggest that school capital investments directly related to classroom infrastructure (HVAC, roofing, classroom renovation etc.) can lead to significant improvements in student test scores and housing prices, with larger effects observed in districts with higher proportions of low-income students. In this paper, we revisit the analysis of Biasi, Lafortune and Schönholzer (2025) and extend their work by exploring state-level heterogeneity in the treatment effect estimates. We first independently replicate their main results in a narrow sense, as our replication exercise reproduces tables and figures in their main text using two different software. We are able to closely replicate their findings using both Stata and R, confirming the robustness of their results. However, we document that their main results consider the full sample of elections rather than imposing a “close election” restriction, which may lead to endogeneity concerns in an RD framework that relies on narrow elections. Next, we expand upon their

analysis to ask whether and how much the effects of school capital investments vary across states. We use their stacked DRD framework to estimate state-specific treatment effects on capital expenditures per pupil, student test scores, and housing price index. Although our results largely corroborate the findings of Biasi, Lafortune and Schönholzer (2025), we highlight some new findings. Our findings reveal significant heterogeneity in the effects of school capital investments across states. Although most states exhibit positive effects of passing a school levy bond on the capital expenditures per pupil, the effect on student test scores and housing prices varies widely. Some states show substantial positive effects, while others exhibit negligible or even negative effects. These variations may be attributed to differences in state-level policies, implementation strategies, and local economic conditions, as well as pre-existing disparities in school infrastructure, which we argue need to be considered when evaluating the effectiveness of school capital investments.

Rest of the paper is organized as follows. In Section 2, we describe the data and methodology used in our replication exercise. In Section 3, we present the results of our replication of the main findings of Biasi, Lafortune and Schönholzer (2025). In Section 4, we extend their analysis by exploring state-level heterogeneity in treatment effects. Finally, in Section 5, we conclude with a discussion of our findings and their implications.

2 Data and Methodology

2.1 Data

Our analysis relies on the national panel of school capital elections and outcomes constructed by Biasi, Lafortune and Schönholzer (2025). They assemble a unified database of school-bond referenda from administrative records and public information requests, covering local bond elections in 41 U.S. states between 1995 and 2017¹. Further, they link these elections to annual district-level finances from the NCES Annual Survey of School System Finances and

¹After sample restrictions, their data comprises of 29 states.

the Census of Governments, student achievement data from the Stanford Education Data Archive (SEDA), and housing price indices from Contat and Larson (2024). This rich dataset allows them to study the fiscal, academic, and housing market responses to school capital investments authorized through local bond elections. Our analysis uses the cleaned master files in their replication package, and we impose all of their sample restrictions as well as some additional restrictions due to limited sample size for certain states².

2.2 Empirical Framework

We follow Biasi, Lafortune and Schönholzer (2025) and exploit quasi-random variation from close bond elections using a dynamic regression discontinuity design. Let m_{jc} denote the vote margin in district j and election cohort c , defined as the difference between the fraction of “yes” votes and the required approval threshold in that state. A bond is authorized if and only if $m_{jc} \geq 0$. Under standard RD assumptions, districts where a bond barely passes ($m_{jc} \gtrsim 0$) provide a valid counterfactual for districts where the bond barely fails ($m_{jc} \lesssim 0$), after flexibly controlling for the running variable. The outcomes Y_{jct} are capital expenditures per pupil, standardized test scores, and the house price index, measured at the district-year level.

Because districts can propose and vote on multiple bonds over time, Biasi, Lafortune and Schönholzer (2025) do not estimate a static, single-election RD. Instead, they develop a *stacked dynamic RD* (DRD) that treats each bond election as a separate “cohort” and follows districts in an event-time window around that election. For each cohort c , they keep up to five pre-election years and ten or more post-election years for districts that place a measure on the ballot in year c and either barely pass or barely fail. Finally, they “stack” all cohorts into one large panel where an observation is indexed by (j, c, t) —district j , cohort c , and calendar year t —and define event time $\tau = t - c$.

In this stacked panel, the main specification relates outcomes to a set of event-time

²Our final analysis consists 24 out of 29 states analysed by Biasi, Lafortune and Schönholzer (2025)

indicators interacted with bond authorization, while controlling for flexible functions of the vote margin and for each district’s bond history. An OLS version of the estimating equation, as presented in Biasi, Lafortune and Schönholzer (2025), is

$$Y_{jct} = \alpha_{jc} + \gamma_{s(j)t} + \sum_{\tau \neq 0} \left[\beta_{\tau} \mathbf{1}\{\tau = t - c\} \times \mathbf{1}\{m_{jc} \geq 0\} + \phi_{\tau} M_{jct}^{(\tau)} + g_c(m_{jc}) \right] + u_{jct},$$

where α_{jc} are district-by-cohort fixed effects, $\gamma_{s(j)t}$ are state-by-year fixed effects, and $g_c(m_{jc})$ denotes cohort-specific polynomials in the vote margin. The indicators $M_{jct}^{(\tau)}$ summarize whether district j proposed or passed other bonds τ years before or after the focal election, so that treated and control districts are compared conditional on their bond histories. The coefficients β_{τ} trace out dynamic treatment effects from five years before to ten or twelve years after authorization with pre-treatment coefficients serve as event-study checks for RD validity. Following Biasi, Lafortune and Schönholzer (2025), all regressions are estimated by OLS, weighting by baseline enrollment or the number of test takers for test-score outcomes, and standard errors are clustered at the district level.

In our replication, we implement this stacked DRD specification in Stata using the original Stata code provided by Biasi, Lafortune and Schönholzer (2025) and port it to R to verify that we can reproduce their tables and figures. For our state-level extension, we use the same estimator and the event-time window, and the same controls for bond histories and vote-margin polynomials, but allow the dynamic treatment effects to vary across states by estimating the model separately for each state.

3 Replication of Main Results

3.1 First Stage

Table 1 presents our replication of the first-stage effects of bond authorization on school expenditures reported in Table II of Biasi, Lafortune and Schönholzer (2025). Our Stata replication reproduces the original estimates exactly, matching both point estimates and standard errors across all three expenditure categories. Our independent R implementation yields nearly identical results, with trivial differences in some cases. The first-stage results confirm that bond passage leads to substantial increases in capital expenditures in years 1–5 of around \$325 per pupil, followed by a decline in years 6–8 as projects near completion, consistent with the dynamics of school construction.

Table 1: Replication of BLS Table II: First Stage Effects on Expenditures

Panel A. Capital expenditure			
Estimate	BLS	Stata	R
1-5 years	325*** (93)	325*** (93)	327*** (93)
6-8 years	-239** (104)	-239** (104)	-207* (106)
9-10 years	2 (110)	2 (110)	46 (110)
Panel B. Current expenditure			
Estimate	BLS	Stata	R
1-5 years	7 (30)	7 (30)	-6 (30)
6-8 years	-32 (47)	-32 (47)	-42 (47)
9-10 years	-64 (51)	-64 (51)	-76 (52)
Panel C. Other non-instructional services			
Estimate	BLS	Stata	R
1-5 years	3 (4)	3 (4)	3 (4)
6-8 years	7 (7)	7 (7)	7 (7)
9-10 years	10 (8)	10 (8)	10 (9)

Notes: Table compares point estimates reported in Table II of Biasi, Lafortune and Schönholzer (2025) (in “BLS” column), estimates produced using the original Stata replication code (in “Stata” column) and our replication code in R (in “R” column). Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.2 Mean Effects

Table 2 replicates the reduced-form effects on test scores and house prices from Table III of Biasi, Lafortune and Schönholzer (2025). Across all four panels, overall test scores, math, ELA³, and the house price index, our Stata and R estimates are virtually indistinguishable from the published results. The independent replication confirms that bond authorization generates economically and statistically significant improvements in student achievement that persist and grow over time, alongside meaningful increases in local house prices.

Table 2: Replication of BLS Table III: Mean Effects on Test Scores and House Prices

Panel A. Test scores			
Estimate	BLS	Stata	R
1-4 years	0.051*** (0.014)	0.051*** (0.014)	0.051*** (0.014)
5-8 years	0.096*** (0.023)	0.096*** (0.023)	0.096*** (0.023)
9-12 years	0.091*** (0.029)	0.091*** (0.029)	0.091*** (0.030)
Panel B. Math			
Estimate	BLS	Stata	R
1-4 years	0.054*** (0.018)	0.054*** (0.018)	0.054*** (0.018)
5-8 years	0.096*** (0.028)	0.096*** (0.028)	0.096*** (0.028)
9-12 years	0.068** (0.033)	0.068** (0.033)	0.068** (0.033)
Panel C. ELA			
Estimate	BLS	Stata	R
1-4 years	0.050*** (0.014)	0.050*** (0.014)	0.050*** (0.014)
5-8 years	0.102*** (0.022)	0.102*** (0.022)	0.102*** (0.022)
9-12 years	0.115*** (0.030)	0.115*** (0.030)	0.115*** (0.030)
Panel D. House Price Index			
Estimate	BLS	Stata	R
1-4 years	2.397* (1.287)	2.397* (1.287)	2.397* (1.291)
5-8 years	6.783*** (2.047)	6.783*** (2.047)	6.783*** (2.055)
9-12 years	4.822** (1.927)	4.822** (1.927)	4.822** (1.934)

Notes: Table compares point estimates reported in Table III of Biasi, Lafortune and Schönholzer (2025) (in “BLS” column), estimates produced using the original Stata replication code (in “Stata” column) and our replication code in R (in “R” column). Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

³ELA stands for english language arts.

3.3 Distribution of Elections

Figure 1 displays the distribution of vote margins in the sample used to produce Tables II and III in Biasi, Lafortune and Schönholzer (2025), where the margin is defined as the difference between the share of "yes" votes and the state-mandated approval threshold. The distribution spans from approximately -40% to $+50\%$, suggesting that the analysis includes elections decided by substantial margins rather than being limited to "close elections". This raises an important identification concern: the validity of the regression discontinuity design fundamentally relies on the assumption that elections near the threshold are quasi-random, with winners and losers being otherwise comparable⁴. When elections far from the cutoff are included, this local randomization assumption becomes tenuous, as districts with landslide victories or defeats likely differ systematically in ways that may confound the causal interpretation of the results. The lack of a bandwidth restriction in the main analysis thus warrants careful consideration when interpreting the estimated treatment effects.

4 State-level Extension

We now extend work done by Biasi, Lafortune and Schönholzer (2025) to examine state-level heterogeneity in treatment effects. While the national-level estimates provide valuable insights into the average impact of school capital investments, they may mask substantial variation across states. To investigate this, we re-estimate the stacked dynamic RD specification separately for each state with sufficient sample size, focusing on three key outcomes: capital expenditures per pupil, standardized test scores, and house price indices.

⁴Although Biasi, Lafortune and Schönholzer (2025) considers concerns regarding manipulation of the running variable by troublesome removing states, observing lack of covariate discontinuities, and balancing on observables, the concern with respect to unobservables still remains for elections away from the margin.

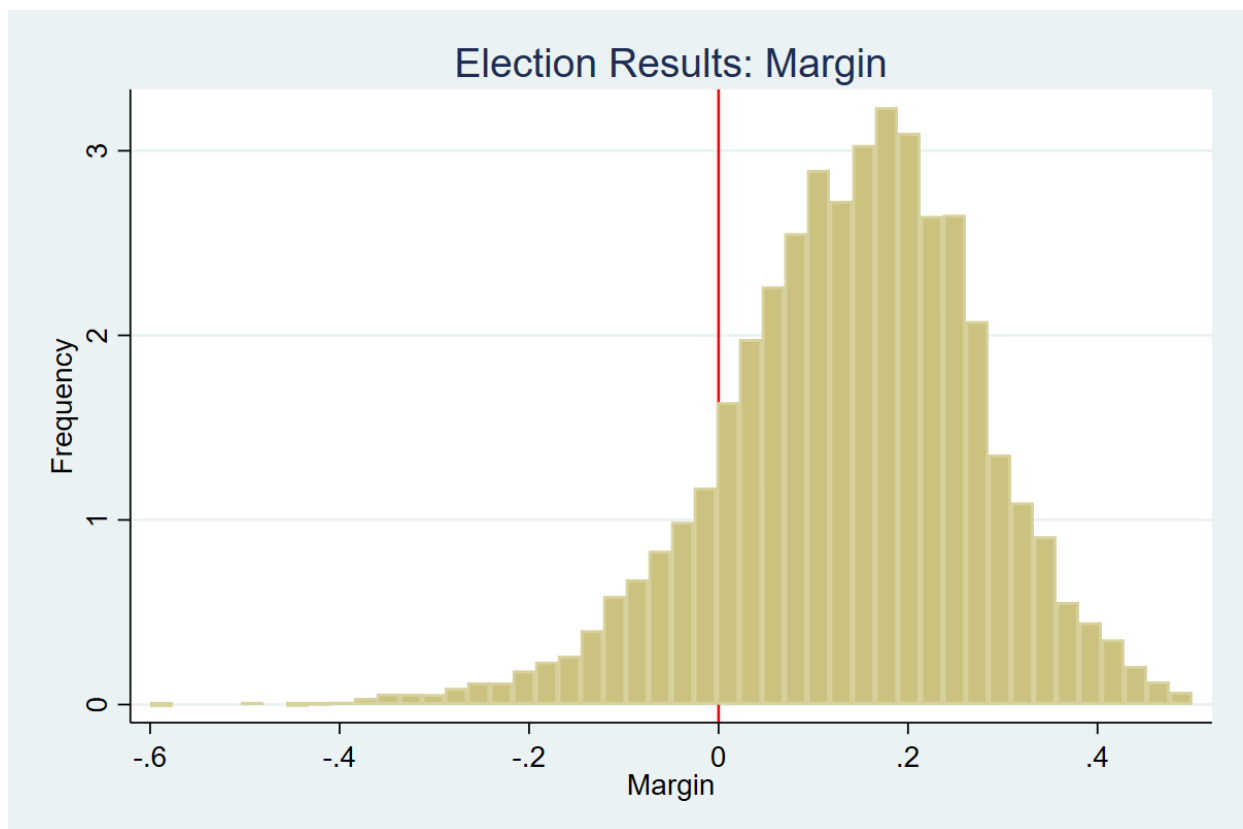


Figure 1: School Bond Election: Distribution of Vote Margins

4.1 First-Stage Effects on Capital Spending

Figure 2 and Table 3 reveal considerable heterogeneity in how bond passage translates into capital spending across states. In the short run (years 1–5), states such as Oregon (\$2,482 per pupil), Pennsylvania (\$2,606), Delaware (\$1,853), and West Virginia (\$1,543) exhibit large and statistically significant increases in capital expenditures following bond authorization. California, Idaho, Indiana, Michigan, Minnesota, Ohio, Texas, and Wisconsin also show substantial positive effects ranging from \$670 to \$1,432 per pupil. However, several states display no discernible first-stage effect: Connecticut shows a large negative coefficient (-\$2,350), though imprecisely estimated, while Arkansas, Iowa, New York, and North Carolina show small or statistically insignificant effects. This variation suggests that bond authorization does not uniformly translate into spending increases even in the short term after passage.

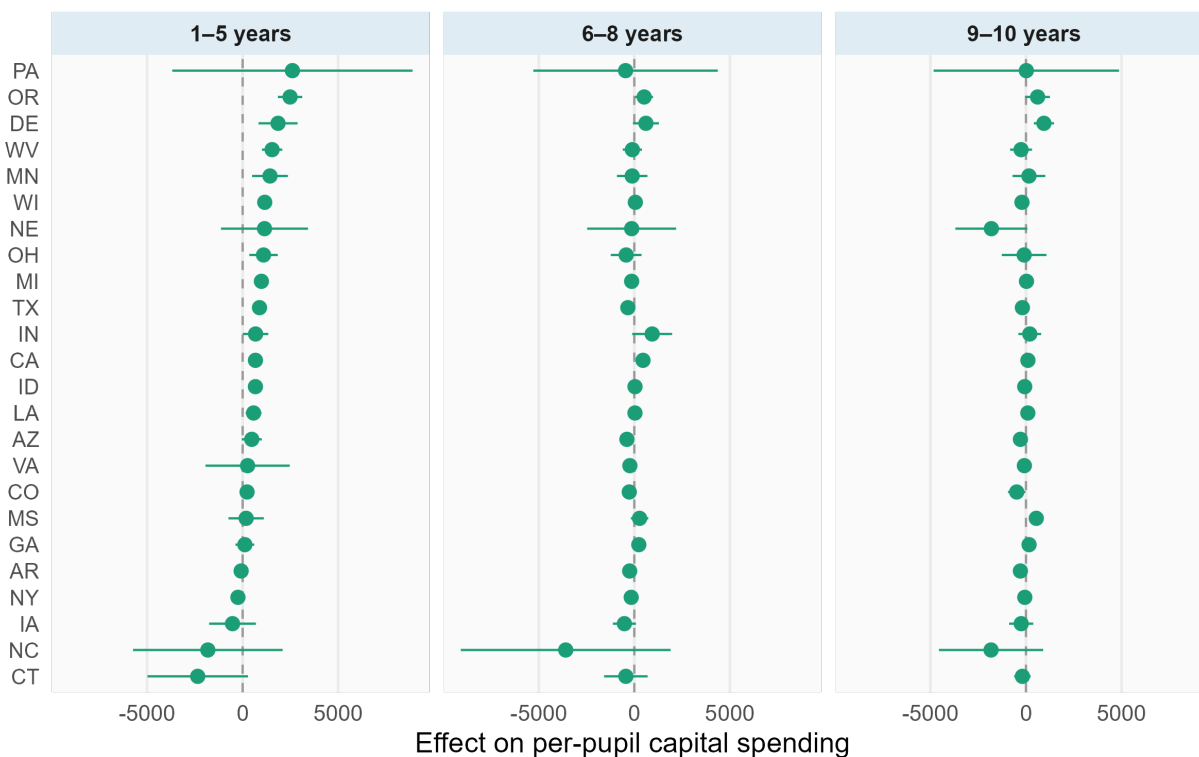


Figure 2: Effect of School Levy on Capital Expenditures per Pupil by State

By the medium term (years 6–8), many states experience a decline in capital spending relative to the initial surge, consistent with the completion of major construction projects.

Arizona, Arkansas, Colorado, Iowa, and Texas show significant negative coefficients in this window, indicating that capital expenditures fall below pre-election levels. Interestingly, some states maintain elevated spending: California (\$456), Oregon (\$515), and Delaware (\$607) continue to exhibit positive effects, suggesting either more gradual implementation or sustained investment in ongoing facility improvements. The long-run effects (years 9–10) are generally smaller in magnitude and often statistically insignificant, although Delaware and Mississippi stand out with persistent positive impacts, while Arizona, Arkansas, Colorado, Iowa, Nebraska, and Wisconsin show significant declines.

4.2 Reduced-Form Effects on Test Scores

The variation in test score effects by states is shown in Figure 3 and Table 4. While Biasi, Lafortune and Schönholzer (2025) find positive average effects nationally, our state-level decomposition reveals that this aggregate pattern conceals substantial heterogeneity, including both large positive and large negative estimates. In the short run (years 1–4), only Delaware (0.295 SD) and Louisiana (0.268 SD) exhibit statistically significant positive effects. Interestingly, several states display negative short-run effects: Indiana (-0.213 SD), Virginia (-0.531 SD), Colorado (-0.120 SD), and West Virginia (-0.938 SD), although most estimates are imprecise. The medium-term effects (years 5–8) show similarly mixed patterns. Arkansas (0.050 SD), California (0.058 SD), Ohio (0.111 SD), and Iowa (0.518 SD) exhibit positive and significant impacts, with Iowa’s estimate being particularly large. North Carolina’s effect remains positive and significant (0.978 SD). However, Idaho (-0.100 SD), Michigan (-0.083 SD), Mississippi (-0.243 SD), Pennsylvania (-0.590 SD), and Virginia (-0.348 SD) all show significant negative effects in this window. The negative estimates in Pennsylvania and Virginia are especially noteworthy given their magnitude. By the long run (years 9–12), Arkansas (0.080 SD), California (0.054 SD), Colorado (0.120 SD), Ohio (0.178 SD), and Texas (0.075 SD) display modest but statistically significant positive effects, while Pennsylvania (-0.687 SD), Virginia (-0.205 SD), West Virginia (-1.721 SD), and Iowa (-0.284 SD) show persistent

negative impacts.

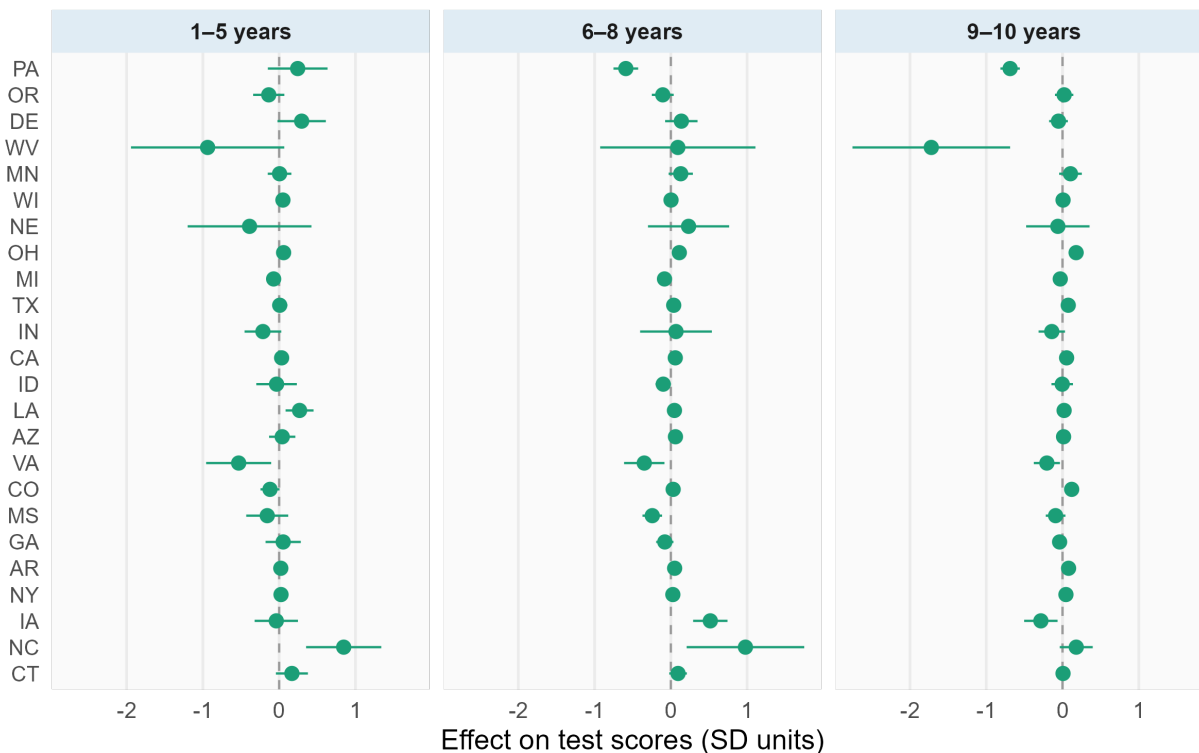


Figure 3: Effect of School Levy on Test Scores by State

This heterogeneity raises important questions about the mechanisms underlying the national average effects. States with negative test score responses may differ in the types of projects funded, the quality of implementation, or the extent to which construction disrupts instruction. Alternatively, these patterns could reflect pre-existing differences in school infrastructure quality: states with relatively better facilities may see smaller gains or even declines if resources are diverted from instruction, while states with deteriorating infrastructure may benefit more from capital improvements.

4.3 Housing Market Responses

Figure 4 and Table 5 show that house price effects are similarly heterogeneous. In the short run (years 1–4), Louisiana exhibits a large positive effect (20.90 points on the HPI), while Virginia shows a marginally significant increase of 37.84 points. However, Idaho displays

a substantial and significant *negative* effect (-22.81 points), and Arkansas shows a smaller negative coefficient (-3.12 points). Most other states exhibit statistically insignificant short-run price responses, suggesting that housing markets may not immediately capitalize the benefits of school capital investments in all contexts.

The medium-term effects (years 5–8) reveal continued divergence. Louisiana (7.88 points), Minnesota (10.61 points), and Mississippi (10.00 points) show positive and significant house price increases, indicating that markets in these states eventually recognize and capitalize the improvements in school quality. In stark contrast, Connecticut (-5.23 points), Idaho (-20.92 points), Michigan (-6.54 points), and North Carolina (-44.78 points) exhibit significant negative effects, with North Carolina's estimate being especially large. The negative house price effects could reflect several mechanisms: migration responses, tax capitalization if property taxes rise to service the bonds, or signaling effects if bond passage reveals negative information about district finances or infrastructure conditions.

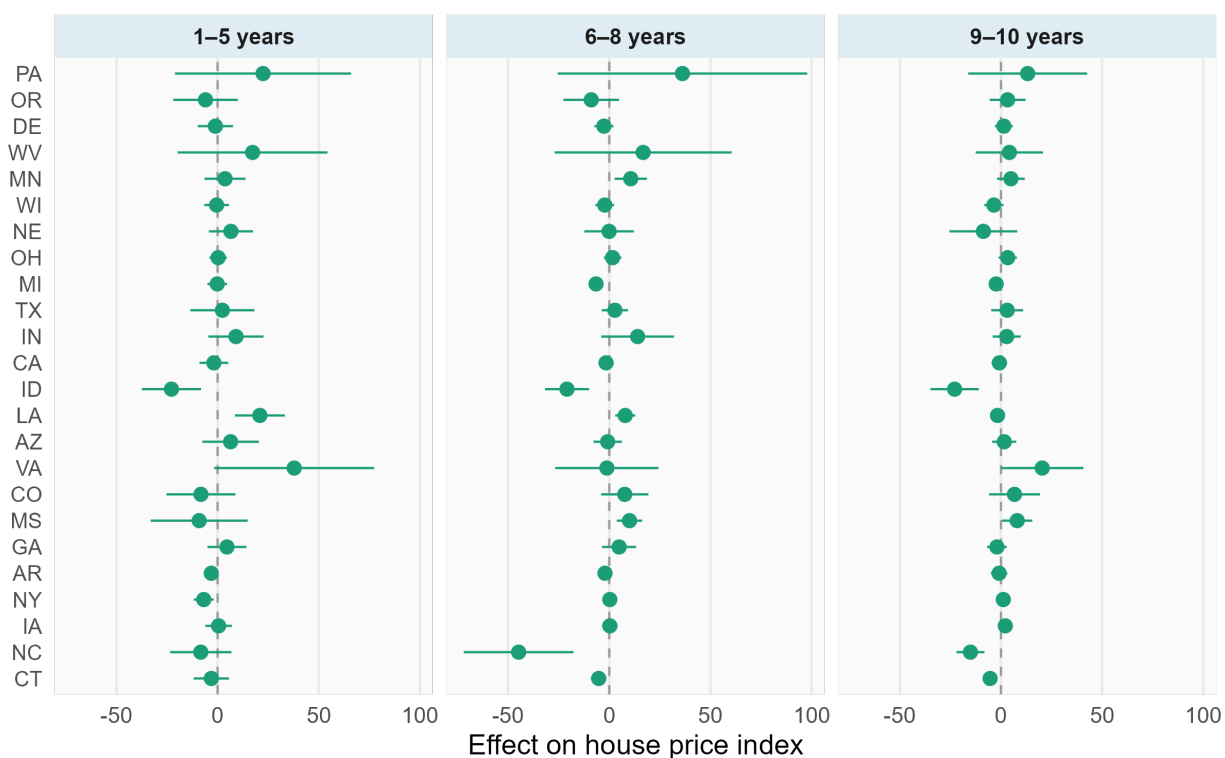


Figure 4: Effect of School Levy on Housing Price Index by State

By the long run (years 9–12), house price effects are generally smaller and less precisely estimated. Connecticut (-5.37 points), Idaho (-22.90 points), and North Carolina (-15.09 points) maintain significant negative effects, while Iowa (2.14 points), Mississippi (8.01 points), and Virginia (20.38 points) show positive impacts. The persistence of negative effects in some states suggests that the housing market response is not merely a short-term adjustment but reflects sustained changes in neighborhood desirability or fiscal conditions.

5 Conclusion

This paper replicates and extends the analysis of Biasi, Lafortune and Schönholzer (2025), who study the effects of school capital investments on student outcomes and house prices using a comprehensive dataset of bond elections across U.S. states. Our replication exercise confirms the robustness of their main findings: using both the original Stata code and an independent R implementation, we reproduce their first-stage and reduced-form estimates with high fidelity. However, our examination of the vote margin distribution reveals an important caveat. The sample used in the main analysis includes elections with margins ranging from approximately -40% to $+50\%$, indicating that many observations are far from the discontinuity threshold. Future work may investigate whether the main results are robust to imposing stricter bandwidth restrictions that limit the sample to truly “close elections”. In this paper, although we replicate the main tables II and III from Biasi, Lafortune and Schönholzer (2025)⁵, we do not replicate sections that differentiate students by income levels and elections by project types. These extensions can be explored in future work to see if the results hold in those dimensions as well.

Our paper documents state-level heterogeneity in treatment effects that is masked by the national average estimates. While Biasi, Lafortune and Schönholzer (2025) find positive aggregate effects on test scores and house prices, our state-by-state decomposition reveals considerable variation—including states with large positive effects, null effects, and even

⁵We also successfully reproduce figures I, II and III in Biasi, Lafortune and Schönholzer (2025).

significant negative effects. This heterogeneity has important implications for policy and future research. First, it suggests that the effectiveness of school capital investments may depend on local context, including pre-existing infrastructure quality, implementation capacity, and community characteristics. Policies designed based on national average effects may be misguided in states where the returns to capital investment differ substantially from the national mean. Second, the presence of negative effects in some states warrants further investigation. Potential explanations include construction disruptions, crowd-out of other productive spending, tax capitalization effects, or differences in how bonds are used across states. Understanding the sources of this variation is essential for designing more effective school finance policies.

Future research should explore the determinants of the observed heterogeneity. Key questions include: Do states with worse baseline infrastructure benefit more from capital investments? Do differences in labor market conditions, construction costs, or procurement practices explain variation in spending efficiency? How do state-level policies regarding school finance equalization or capital project oversight affect outcomes? Answering these questions will require linking our state-specific estimates to detailed data on district characteristics, project attributes, and institutional features. Such analysis could help identify the conditions under which school capital investments are most likely to improve student outcomes and neighborhood quality, thereby informing more efficient allocation of scarce public resources.

Data and Code Availability

All data used in this paper is from Biasi, Lafortune and Schönholzer (2025). Replication code will be deposited to the Journal of Applied Econometrics (JAE) Data Archive, in accordance with the journal’s data and code policy.

References

- Abott, Carolyn, Vladimir Kogan, Stéphane Lavertu, and Zachary Peskowitz.** 2020. “School district operational spending and student outcomes: Evidence from tax elections in seven states.” *Journal of Public Economics*, 183: 104142.
- Baron, E Jason.** 2022. “School spending and student outcomes: Evidence from revenue limit elections in Wisconsin.” *American Economic Journal: Economic Policy*, 14(1): 1–39.
- Biasi, Barbara, Julien Lafortune, and David Schönholzer.** 2025. “What Works and for Whom? Effectiveness and Efficiency of School Capital Investments Across the U.S.” *The Quarterly Journal of Economics*.
- Cellini, Stephanie R, Fernando Ferreira, and Jesse Rothstein.** 2010. “The value of school facility investments: Evidence from a dynamic regression discontinuity design.” *The Quarterly Journal of Economics*, 125(1): 215–261.
- Contat, Justin, and William D Larson.** 2024. “A flexible method of housing price index construction using repeat-sales aggregates.” *Real Estate Economics*, 52(6): 1551–1583.
- Martorell, Paco, Kevin Stange, and Isaac McFarlin Jr.** 2016. “Investing in schools: capital spending, facility conditions, and student achievement.” *Journal of Public Economics*, 140: 13–29.
- Neilson, Christopher A, and Seth D Zimmerman.** 2014. “The effect of school construction on test scores, school enrollment, and home prices.” *Journal of Public Economics*, 120: 18–31.

Appendix

Table 3: Effect of School Capital Levies on Capital Spending by State

State	Capital spending (per-pupil dollars)		
	1–5 years	6–8 years	9–10 years
AZ	478.2 (267.3)	-388.3*** (94.8)	-291.9*** (82.6)
AR	-77.9 (125.9)	-242.2** (114.0)	-293.4 (186.9)
CA	670.9*** (134.3)	455.9*** (79.8)	99.8 (95.8)
CO	231.4 (190.0)	-264.8 (175.9)	-486.8** (232.0)
CT	-2350.4* (1341.6)	-442.0 (581.8)	-187.7 (219.5)
DE	1853.0*** (521.7)	607.2* (347.0)	937.7*** (269.5)
GA	116.7 (250.9)	235.3** (114.1)	160.4 (143.4)
ID	669.5*** (143.2)	36.2 (75.7)	-67.9 (114.5)
IN	677.1** (340.1)	934.5* (531.9)	198.2 (303.5)
IA	-529.0 (624.7)	-519.8* (308.6)	-250.6 (321.7)
LA	573.0*** (210.8)	34.5 (69.1)	96.3 (97.5)
MI	981.9*** (172.5)	-140.0 (93.6)	24.9 (101.5)
MN	1432.8*** (477.9)	-111.2 (406.5)	153.9 (437.1)
MS	183.7 (472.1)	276.1 (233.4)	539.4*** (188.5)
NE	1145.6 (1162.1)	-138.6 (1187.3)	-1811.8* (961.3)
NY	-245.5 (194.4)	-162.9 (128.2)	-63.0 (139.9)
NC	-1820.0 (1998.0)	-3590.2 (2801.1)	-1828.3 (1391.6)
OH	1094.1*** (379.9)	-426.4 (410.9)	-97.5 (596.4)
OR	2482.2*** (325.4)	515.4** (237.4)	607.5* (328.9)
PA	2605.8 (3208.0)	-458.7 (2463.1)	16.7 (2475.1)
TX	883.5*** (163.4)	-338.2** (149.3)	-188.7 (136.4)
VA	259.3 (1124.9)	-231.9 (186.1)	-85.9 (192.9)
WV	1542.5*** (272.4)	-101.8 (256.2)	-257.5 (293.1)
WI	1156.3*** (185.1)	53.0 (97.9)	-211.5** (103.1)

Notes: Entries report state-specific treatment effects of passing a school capital levy on per-pupil capital spending over the indicated post-election horizons. Standard errors in parentheses. Horizon bins follow the specification in Biasi, Lafortune, and Schönholzer (2025). *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4: Effect of School Capital Levies on Test Scores by State

State	Test scores (SD units)		
	1–4 years	5–8 years	9–12 years
AZ	0.040 (0.088)	0.061 (0.047)	0.014 (0.038)
AR	0.022 (0.027)	0.050* (0.027)	0.080*** (0.029)
CA	0.032 (0.033)	0.058*** (0.015)	0.054*** (0.019)
CO	-0.120* (0.064)	0.031 (0.036)	0.120*** (0.031)
CT	0.167 (0.107)	0.094 (0.059)	0.007 (0.038)
DE	0.295* (0.162)	0.138 (0.108)	-0.053 (0.063)
GA	0.053 (0.118)	-0.079 (0.058)	-0.038 (0.030)
ID	-0.034 (0.136)	-0.100** (0.051)	-0.004 (0.072)
IN	-0.213* (0.123)	0.068 (0.240)	-0.141 (0.088)
IA	-0.038 (0.145)	0.518*** (0.115)	-0.284** (0.112)
LA	0.268*** (0.094)	0.047 (0.037)	0.021 (0.031)
MI	-0.072** (0.032)	-0.083*** (0.028)	-0.030* (0.018)
MN	0.006 (0.079)	0.130 (0.081)	0.104 (0.076)
MS	-0.156 (0.141)	-0.243*** (0.066)	-0.091 (0.066)
NE	-0.388 (0.414)	0.233 (0.272)	-0.061 (0.212)
NY	0.025 (0.024)	0.026 (0.018)	0.045* (0.023)
NC	0.846*** (0.252)	0.978** (0.393)	0.180 (0.110)
OH	0.058 (0.044)	0.111*** (0.031)	0.178*** (0.049)
OR	-0.136 (0.104)	-0.106 (0.073)	0.021 (0.061)
PA	0.243 (0.200)	-0.590*** (0.083)	-0.687*** (0.065)
TX	0.008 (0.051)	0.038 (0.034)	0.075** (0.036)
VA	-0.531** (0.218)	-0.348*** (0.135)	-0.205** (0.087)
WV	-0.938* (0.513)	0.092 (0.519)	-1.721*** (0.527)
WI	0.050 (0.039)	0.003 (0.035)	0.007 (0.021)

Notes: Entries report state-specific treatment effects of passing a school capital levy on test scores (in within-state standard deviation units) over the indicated post-election horizons. Standard errors in parentheses. Horizon bins follow the specification in Biasi, Lafortune, and Schönholzer (2025). *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5: Effect of School Capital Levies on House Prices by State

State	House price index		
	1–4 years	5–8 years	9–12 years
AZ	6.43 (7.12)	-0.80 (3.57)	1.63 (3.04)
AR	-3.12* (1.80)	-2.13 (1.63)	-0.91 (2.07)
CA	-1.85 (3.64)	-1.59 (1.93)	-0.77 (1.96)
CO	-8.22 (8.69)	7.64 (5.96)	6.68 (6.42)
CT	-3.10 (4.45)	-5.23*** (1.93)	-5.37*** (1.22)
DE	-1.10 (4.45)	-2.65 (2.41)	1.43 (2.25)
GA	4.63 (4.92)	4.86 (4.30)	-2.00 (2.50)
ID	-22.81*** (7.47)	-20.92*** (5.55)	-22.90*** (6.10)
IN	9.08 (6.94)	14.00 (9.18)	2.85 (3.54)
IA	0.48 (3.35)	0.28 (1.97)	2.14* (1.20)
LA	20.90*** (6.29)	7.88*** (2.53)	-1.70 (1.71)
MI	-0.19 (2.48)	-6.54*** (1.71)	-2.37 (1.59)
MN	3.68 (5.16)	10.61*** (4.07)	4.92 (3.50)
MS	-9.10 (12.23)	10.00*** (3.21)	8.01** (3.82)
NE	6.58 (5.57)	-0.08 (6.23)	-8.69 (8.55)
NY	-6.84*** (2.52)	0.27 (0.97)	1.15 (1.00)
NC	-8.32 (7.72)	-44.78*** (13.83)	-15.09*** (3.53)
OH	0.27 (2.19)	1.65 (2.20)	3.33 (2.31)
OR	-5.98 (8.15)	-8.90 (7.01)	3.28 (4.51)
PA	22.47 (22.19)	36.16 (31.46)	13.18 (15.00)
TX	2.37 (8.07)	2.76 (3.32)	3.07 (4.02)
VA	37.84* (20.18)	-1.20 (13.02)	20.38** (10.38)
WV	17.27 (18.90)	16.73 (22.31)	4.15 (8.48)
WI	-0.49 (3.11)	-2.25 (2.38)	-3.50 (2.44)

Notes: Entries report state-specific treatment effects of passing a school capital levy on local house price index over the indicated post-election horizons. Standard errors in parentheses. Horizon bins follow the specification in Biasi, Lafortune, and Schönholzer (2025). *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6: Mapping of FIPS Codes to States in the State-Level Analysis

FIPS code	State abbreviation	State name
04	AZ	Arizona
05	AR	Arkansas
06	CA	California
08	CO	Colorado
09	CT	Connecticut
10	DE	Delaware
13	GA	Georgia
16	ID	Idaho
18	IN	Indiana
19	IA	Iowa
22	LA	Louisiana
26	MI	Michigan
27	MN	Minnesota
28	MS	Mississippi
31	NE	Nebraska
36	NY	New York
37	NC	North Carolina
39	OH	Ohio
41	OR	Oregon
42	PA	Pennsylvania
48	TX	Texas
51	VA	Virginia
54	WV	West Virginia
55	WI	Wisconsin